AITS
ALL INDIA TEST SERIES

Dropper JEE Advanced

AJPM9.2/08 | **Test- 08**

DURATION : 180 Minutes | **DATE : 16/02/2025** | **M.MARKS : 180**

Topics covered

Physics:	Full Syllabus
Chemistry:	Full Syllabus
Mathematics:	Full Syllabus

General Instructions:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **3 hours** duration.
3. The test booklet consists of 54 questions. The maximum marks are **180**.

SECTION-1 (Maximum Marks: 24)

This section contains **EIGHT (08)** Questions.

- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from **0 TO 9, BOTH INCLUSIVE**.
- For each question, enter the correct integer corresponding to the answer using the mouse and the on screen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:

  Full Marks : +3 If **ONLY** the correct integer is entered;
  Zero Marks : 0 In all other cases.
  Negative Marks : –1 In all other cases.

SECTION-2 (Maximum marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme:

  Full Marks : +4 **ONLY** if (all) the correct option(s) is(are) chosen;
  Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen;
  Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct;
  Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option;
  Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
  Negative Marks : –2 In all other cases.

SECTION-3 (Maximum marks: 12)

- This section contains **FOUR (04)** Questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

  Full Marks : +3 **ONLY** if the option corresponding to the correct combination is chosen;
  Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
  Negative Marks : –1 In all other cases.

OMR Instructions:

1. Use blue/black dark ballpoint pens.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. Do not fold or make any stray mark on the Answer Sheet (OMR).

Name of the Students (In CAPITALS) : _____

Roll Number : _____

OMR Bar Code Number : _____

Candidate's Signature : _____ Invigilator's Signature _____

IMPORTANT CONSTANTS

Speed of light in free space,	:	$3.00 \times 10^{8}\ ms^{-1}$
Permeability of free space,	:	$4\pi \times 10^{-7}\ Hm^{-1}$
Permittivity of free space,	:	$8.85 \times 10^{-12}\ Fm^{-1}$
The Planck constant,	:	$6.63 \times 10^{-34}\ Js$
Rest mass of electron,	:	$9.1 \times 10^{-31}\ kg$
Rest mass of proton,	:	$1.67 \times 10^{-27}\ kg$
Molar gas constant,	:	$8.31\ JK^{-1}\ mol^{-1}$
The Avogadro constant,	:	$6.02 \times 10^{23}\ mol^{-1}$
The Boltzmann constant,	:	$1.38 \times 10^{-23}\ JK^{-1}$
Gravitational constant,	:	$6.67 \times 10^{-11}\ N\ m^2kg^{-2}$
Acceleration of free fall	:	$9.8\ ms^{-2}$
Rydberg Constant	:	$1.097 \times 10^{7}\ m^{-1}$
Atomic mass unit	:	$1.67 \times 10^{-27}\ kg$
Charge on proton	:	$1.6 \times 10^{-19}\ C$

IMPORTANT VALUES

$\sqrt{2}$ = 1.414

$\sqrt{3}$ = 1.732

$\sqrt{5}$ = 2.236

π = 3.142

e (Euler's constant) = 2.718

ln 10 = 2.303

$\log_{10}2$ = 0.3010

$\log_{10}3$= 0.4770

$\log_{10}7$ = 0.845

**** Use above values unless otherwise specified in a question.***

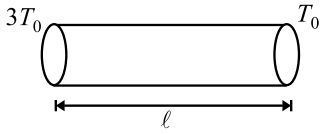
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PART-I (PHYSICS)

SECTION-1

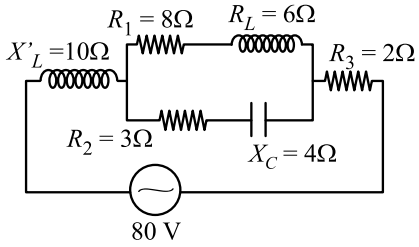
Single Digit Integer Type Questions:

1. Two ends of a rod of uniform cross sectional area are kept at temperatures $3T_0$ and T_0 as shown if the thermal conductivity of rod varies with temperature as. $k = \alpha T$ (where α is a constant and T is temperature). In steady state, if the temperature of the middle section of the rod is xT_0, then the value of x^2 is_____.



2. Two charges $q > 0$ and $-q$ are located along the x-axis at $(-d,0,0)$ and $(d,0,0)$. If R is the radial distance from the origin that a field line emanating from q at an angle $\alpha=60°$, with respect to the positive x-axis intersects with the y-z plane. Then find the value of $\frac{9R^2}{d^2}$.

3. Consider the circuit shown in the figure, the magnitude of current (in A) supplied by the AC voltage source is

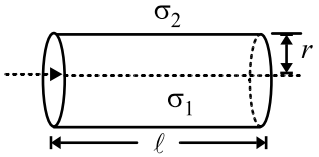


4. Consider a gaseous sample of Li^{2+} ions. All the ions (a total of 10^8) are in 3rd excited state. The radiations from this sample, when ions are transitioning to the ground state, are made to fall on a target sample of work function 95 eV. Assuming that the different possible journeys (of reaching the ground state from the initial state) are equilikely, the best estimate of the number of photoelectrons ejected from the target is $x \times 10^8$. Find the value of $\frac{1}{x}$. (Take ionisation energy of an H atom to be 13.6 eV and assume that all electrons are able to leave metal surface with minimum threshold energy.)

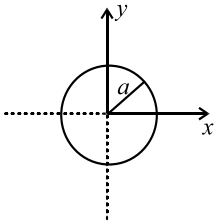
5. A body is projected vertically upward from the surface of earth with a velocity equal to the escape velocity. The radius of earth is R and the acceleration due to gravity on the surface of the earth is g. The time taken by the body to reach to the height h from surface of earth is $\frac{1}{k}\sqrt{\frac{2R}{g}}\left[\left(1+\frac{h}{R}\right)^{3/2}-1\right]$. Find the value of k.

6. A long conductor of circular cross-section has radius r and length l as shown in the figure. The conductivity of the material near the axis is σ_1 and increases linearly with the distance from axis and becomes σ_2 near the surface. The resistance of the conductor if the current enters from the one end and leaves from the other end is $\frac{nl}{\pi r^2(\sigma_1+2\sigma_2)}$.

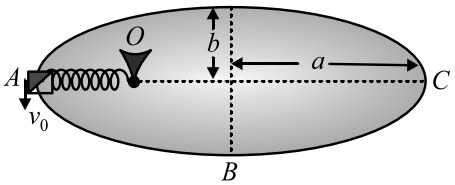
Find the value of n.



7. A circular ring of radius a and resistance R is placed (fixed) in x–y plane with its centre at origin as shown in the figure. A magnetic field $\vec{B}=(\alpha x)\hat{i}+(\beta y)\hat{j}+(\gamma t)\hat{k}$ is switched on at $t = 0$. If the force acting on the ring is $\frac{\pi^2 a^4 \gamma(n\alpha+m\beta)}{R}$. Find the value of $(n + m)$.



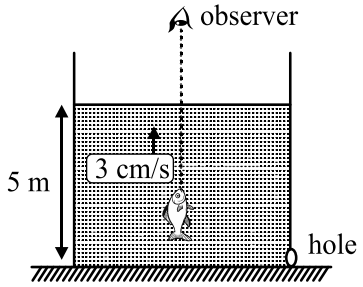
8. A slider can slide on a smooth vertical elliptical guide as shown in the figure. The fixed point O is located at one focus of the elliptical guide. One end of spring is connected at O and the other end is connected to the slider. The spring is unstretched when the slider is at A. The speed $v_0 = 2\sqrt{3}$ m/s is given to the slider at A is such that the speed approaches to zero at C. The semi-major and semi-minor axes are a = 3 m and b = 2 m respectively. The acceleration due to gravity is g = 10 m/s^2. The speed of the slider at B is (in m/sec)



SECTION-2

One or More Than One Correct Type Questions:

9. A tank having cross sectional area A has a hole at the bottom of area of cross section $A_1 = \frac{A}{1000}$. Bottom of the tank is a plane mirror. The tank contains water of refractive index $\frac{4}{3}$. At the instant, when height of the water in the tank is 5 m, a fish is rising vertically in the tank with a velocity 3 cm/s toward the surface. (Assume steady flow and take g =10 m/s^2)



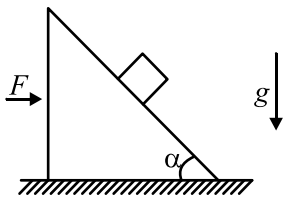
(A) The velocity with which free surface is falling down inside container is 1 cm/s.

(B) The velocity with which free surface is falling down inside container is 2 cm/s.

(C) The velocity of the fish as observed by the observer looking directly at the fish is 2 cm/s.

(D) The velocity of the image of the fish (formed by plane mirror) as observed by the observer is 2.5 cm/sec vertically downwards

10. A weightless wedge is placed on a horizontal surface. There is a body of mass m on the slope of the wedge. A horizontal force F is applied on the wedge as shown. It is observed that, the block moves up along the incline with respect to wedge. There is no friction. Acceleration due to gravity is g . The slope of the wedge makes an angle α from the horizontal



(A) The acceleration of the body is

$$a = \sqrt{\left(\frac{F}{m\sin\alpha}\right)^2 + g^2 - \frac{F}{m}g\cot\alpha}$$

(B) The acceleration of the body is

$$a = \sqrt{\left(\frac{F}{m\sin\alpha}\right)^2 + g^2 - 2\frac{F}{m}g\cot\alpha}$$

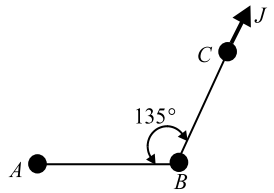
(C) The angle made by the acceleration of the body with the horizontal is

$$\phi = \tan^{-1}\left(\frac{F\sin\alpha - mg\cos\alpha}{F\cos\alpha}\right)$$

(D) The angle made by the acceleration of the body with the horizontal is

$$\phi = \tan^{-1}\left(\frac{F\cos\alpha - mg\sin\alpha}{F\sin\alpha}\right)$$

11. Three identical particles *A, B* and *C* lie on a smooth horizontal table. Light inextensible strings which are just taut connect *AB* and *BC* and ∠*ABC* is 135°. An impulse *J* is imparted to the particle *C* in the direction *BC* . Mass of each particle is *m*. Choose the correct options.



(A) Speed of *A* just after the impulse imparted is $\frac{\sqrt{2}J}{7m}$

(B) Speed of *B* just after the impulse imparted is $\frac{\sqrt{10}J}{7m}$

(C) Speed of *C* just after the impulse imparted is $\frac{3J}{7m}$

(D) Speed of *A* just after the impulse imparted is $\frac{2J}{7m}$

12. An object of mass m moving with velocity v, explodes suddenly in two parts of mass ratio 1:2. If the direction of motion of two parts makes an angle 60° with each other after explosion and de-Broglie wavelength of the lighter part is half of the de-Broglie wavelength of the heavier part, then

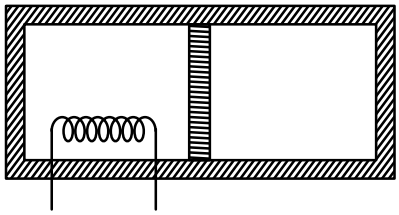
(A) Speed of the lighter part after explosion is $\frac{6v}{\sqrt{7}}$

(B) Speed of the heavier part after explosion is $\frac{3v}{2\sqrt{7}}$

(C) Increase in kinetic energy of the system due to explosion is $\frac{4}{7}mv^2$

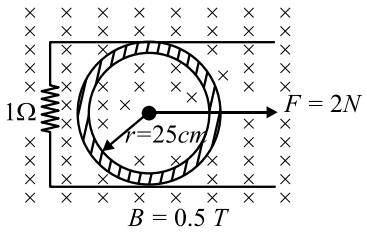
(D) Ratio of kinetic energy of lighter part to that of heavier part is 4:1

13. An insulating cylinder of volume 20 litre is divided into two equal parts by a movable piston made of insulating material. Monoatomic ideal gas is filled at temperature 300 K and pressure 10^5 N/m^2 in each part. Left part is given heat Q joule by using an electric heater such that temperature in the right part becomes 302 K and in the left part becomes $(300 + \Delta T)$ K. (Take $\left(\frac{300}{302}\right)^{3/2} = 0.99$)



(A) The value of ΔT is 12.8 K
(B) The value of Q is 42.6 J
(C) The value of ΔT is 8.10 K
(D) The value of Q is 50.5 J

14. In the shown figure, two parallel rails in a vertical plane of negligible resistance are connected by a resistance R = 1 Ω at one end. A uniform horizontal magnetic field is present (B = 0.5 T). A thin ring of mass m_1=1 kg, radius r =25 cm is fitted over a wooden disc of mass m_1=1 kg. Resistance of ring R' = 4 Ω and there is no friction between the ring and the disc, ring and upper rail. Friction between ring and lower rail is sufficient to support pure rolling of the ring. Horizontal force F = 2 N is applied at centre of the disc. When the terminal velocity is reached

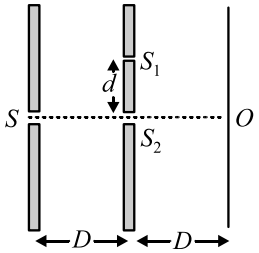


(A) Terminal velocity of disc is 64 m/s
(B) Terminal velocity of disc is 96 m/s
(C) Current flowing in the rails is $8\sqrt{2}$ A
(D) Current flowing in the rails is 8 A

SECTION-3

Single Correct Type Questions:

15. In a *YDSE* experiment the two ideal slits S_1 and S_2 are at a distance d apart, and illuminated by light of wave length λ passing through an ideal source slit S placed on the line through S_2 as shown. The distance between the planes of slits and the source slit is D. A screen is held at a distance D from the plane of the slits. The minimum value of d for which there is a darkness at point O is
(given $D >> d$)

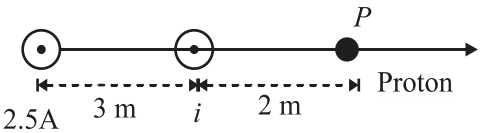


(A) $\sqrt{\frac{3\lambda D}{2}}$
(B) $\sqrt{\lambda D}$
(C) $\sqrt{\frac{\lambda D}{2}}$
(D) $\sqrt{3\lambda D}$

16. A long, straight wire carries a current i. A particle having a positive charge q and mass m, kept at a distance x_0 from the wire is projected towards it with a speed v. Find the minimum separation between the wire and the particle.

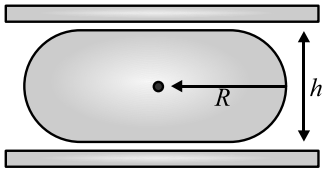
(A) $\frac{x_0}{4}$
(B) $x_0 e^{\frac{-\pi m v}{\mu_0 q i}}$
(C) $x_0 e^{\frac{-2\pi m v}{\mu_0 q i}}$
(D) $-\frac{x_0}{4}$

17. A long thin conductor carries a current 2.5 Amp and another long conductor carries an unknown current i in the same direction as shown in the figure. A proton at point P moving with a velocity 2×10^5 m/s as shown experiences a force of 1.6×10^{-20} N. Then the current i (in ampere), is



(A) 1
(B) 2
(C) 3
(D) 4

18. A mercury drop shaped as a round tablet of radius R and thickness h is placed between two horizontal glass plates. Now a mass m is placed on the upper plate and due to its weight the distance between the plates is diminished n times. The surface tension is T and acceleration due to gravity is g. Find the angle of contact θ. (Assume is $h << R$)



(A) $\pi - \cos^{-1}\left(\frac{2mgh}{\pi R^2 T\left(n^2-1\right)}\right)$

(B) $\pi - \sin^{-1}\left(\frac{2mgh}{\pi R^2 T\left(n^2-1\right)}\right)$

(C) $\pi - \sin^{-1}\left(\frac{mgh}{2\pi R^2 T\left(n^2-1\right)}\right)$

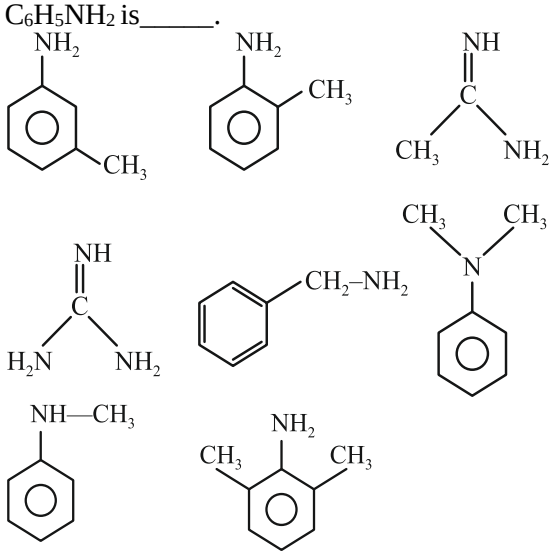
(D) $\pi - \cos^{-1}\left(\frac{mgh}{2\pi R^2 T\left(n^2-1\right)}\right)$

PART-II (CHEMISTRY)

SECTION-1

Single Digit Integer Type Questions:

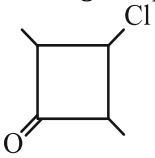
19. The number of species which are more basic than $C_6H_5NH_2$ is_____.



20. A transition metal of 3d-series exhibit its highest oxidation state in its amphoteric oxide as well as in its fluoride.
The group number of this element in long form of periodic table = x.
Sum of ns + (n – 1)d electrons in it = y
The value of |x – y| is _____.

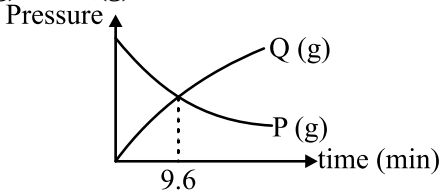
21. Total number of possible stereoisomers for the given formula [Co(en)ClBrI(NH_3)] is x. Then $\frac{x}{2}$ is

22. Consider the following compound



Total number of stereoisomers of this compound is x and number of pair of enantiomers is y.
Find the value of (x + y).

23. For a given 1st order reaction, consider the graph of pressure versus time
$P(g) \rightarrow 2Q(g)$



If half-life of the reaction is x min, then $\sqrt{x}$ is (use log 2 = 0.3 and log 3 = 0.48)

24. How many of the following compounds can react with $NaHCO_3$ and liberate $CO_2(g)$?

(i) Salicylic acid	(ii) Pthalic acid
(iii) Picric acid	(iv) Resorcinol
(v) Carbolic acid	(vi) Aspirin
(vii) Anisole	(viii) Tartaric acid

25. 100 ml of sample of tap water contain 0.192 mg of Mg^{+2}, what is the hardness as parts of $CaCO_3$ per million parts of water?

26. Two bulbs P and Q are emitting monochromatic light of wavelength such that P can just ionize H-atom and Q can just ionize He^+ ion. If power of bulbs P and Q are 60 W and 80 W respectively, then the ratio of number of photons emitted per sec by bulbs P to Q is_____.

SECTION-2

One or More Than One Correct Type Questions:

27. In which one of the following reaction(s), given change does not takes place?

(A) Formic acid $\xrightarrow[H_2SO_4]{\text{conc.}}$ Decarboxylation

(B) Oxalic acid $\xrightarrow[H_2SO_4]{\text{conc.}}$ Decarboxylation

(C) β-keto acid $\xrightarrow{\Delta}$ Decarboxylation

(D) Succinic acid $\xrightarrow{\Delta}$ Decarboxylation

28. Which of the statement(s) is/are true?

(A) The equivalent weight of Na_3PO_4 is M/3 (M is molecular weight)

(B) The highest percentage labelling on oleum bottle is 122.5

(C) The oxidation of iron in $[Fe(NO)(H_2O)_5]SO_4$ is +1

(D) The average oxidation number of iron in Prussian blue is 18/7

29. Consider the following values of I.E. (eV) for elements W and X :

Element	$I.E._1$	$I.E._2$	$I.E._3$	$I.E._4$
W	10.5	15.5	24.9	79.8
X	8	14.8	78.9	105.8

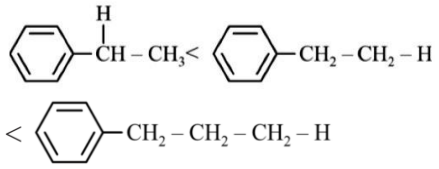
Other two elements Y and Z have outer electronic configuration ns^2np^4 and ns^2np^5 respectively. Then according to given information, which of the following compound(s) is/are not possible?

(A) W_2Y_3 (B) X_2Y_3

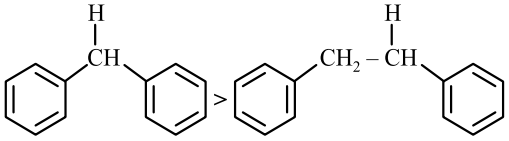
(C) W_2Z_3 (D) XZ_2

30. The correct reactivity order of C—H bond towards photolytic chlorination is/are

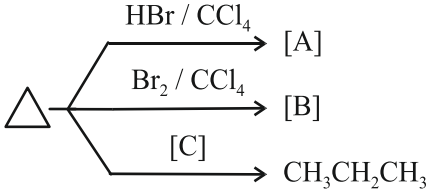
(A) $CH_2=CH-CH_2-H > CH_3CH_2-H > CH_3-H$

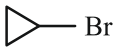
(B) 

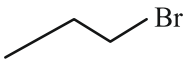
(C) $CH_3CH_2CH_2CH_2-H > CH_3CH_2\underset{\displaystyle H}{\underset{|}{C}}H-CH_3$

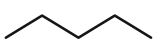
(D) 

31. In the given reaction identify the unknown reagent(s) or product(s)



(A) Compound [B] is 

(B) Compound [A] is 

(C) Compound [B] is Br 

(D) Compound [C] is LAH

32. Select the correct statement(s):

(A) Sucrose contains α – 1, 4 glycosidic linkage

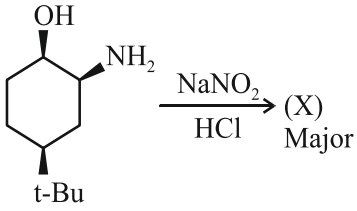
(B) Sucrose contains α – 1, β-2 glycosidic linkage

(C) Maltose contains α – 1, 4 glycosidic linkage

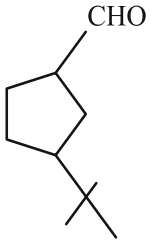
(D) Lactose contains β – 1, 4 glycosidic linkage

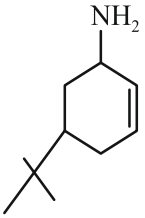
SECTION-3

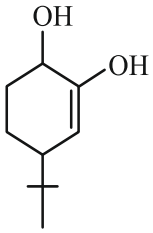
Single Correct Type Questions:

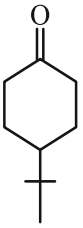
33. 

Product 'X' is

(A) 

(B) 

(C) 

(D) 

34. Which of the following pair is expected to form colourless compounds?

(A) $_{57}La^{2+}$, $_{59}Pr^{3+}$ (B) $_{57}La^{3+}$, $_{71}Lu^{3+}$

(C) $_{60}Nd^{3+}$, $_{61}Pm^{3+}$ (D) $_{63}Eu^{3+}$, $_{65}Tb^{3+}$

35. In the preparation of iron from haematite (Fe_2O_3) by the reduction with carbon,

$Fe_2O_3 + C \rightarrow Fe + CO_2$

How much 80% pure iron may be produced from 120 kg of 90% pure Fe_2O_3?

(A) 94.5 kg (B) 60.48 kg

(C) 116.66 kg (D) 120 kg

36. Identify the CORRECT statements.

(1) NH_4Cl is added along with NH_4OH so that only III[rd] group cations can be precipitated as their hydroxides.

(2) In III[rd] group, in place of NH_4Cl, $(NH_4)_2SO_4$ cannot be used as V[th] group radical(s) will also be precipitated as MSO_4 along with Al^{3+}, Fe^{3+} & Cr^{3+}.

(3) Aqueous solution of ammonium sulphate produces white ppt. with $BaCl_2$ solution which is insoluble in dil. HCl.

(A) 1 and 2 only (B) 2 and 3 only

(C) 1 and 3 only (D) 1, 2 and 3

PART-III (MATHEMATICS)

SECTION-1

Single Digit Integer Type Questions:

37. Let $f(x)=\begin{vmatrix} x & 1 & 1 \\ \sin 2\pi x & 2x^2 & 1 \\ x^3 & 3x^4 & 1 \end{vmatrix}$. If $g(x) + f(x) = 0$.

If $\lambda f(1) g(1) = -2$, then the value of 2λ is

38. If z is a complex number satisfying $|z - 4| = 2$ and the minimum value of $|z|$ is λ and if $y = 2[x] + 3 = 3[x - \lambda]$, then the value of $\frac{[x+y]}{10}$ is (where [.] denotes the greatest integer function)

39. The number of integral solution of the equation $x + y + z + w < 25$ such that $x > -2$, $y > 1$, $z \geq 0$, $w > 3$ is P. Then the value of $\frac{P+145}{3000}$ is

40. The value of $\frac{1}{2}\sec 80° - \frac{\sqrt{3}}{2}\operatorname{cosec} 80°$ is

41. Let M and m be maximum and minimum values of function $f(x)=\frac{x^2-x+1}{x^2+x+1}$ at $x = \alpha$ and $x = \beta$ respectively then value of $\frac{M-3m}{|\alpha-\beta|}$ is equal to

42. If area of the region by two branches of $(y-x)^2 = x^3$ and straight line $x = 1$ is A. Then the value of $5A$ is

43. Mr. Akash lists all the positive divisors of the number $N = (2010)^2$ and selects two divisors from the list then the probability that exactly one of the selected divisors is perfect squares is $\frac{p}{q}$. Then $\frac{2p+3q}{295}$ (where p, q are co-prime)

44. If $\vec{P}_1 = \hat{i} + \hat{j} + \hat{k}$ and $\vec{P}_2 = a\hat{i} + b\hat{j} + c\hat{k}$ where $a, b, c \in \{-2, -1, 0, 1, 2\}$, then number of possible non-zero vectors $\vec{P}_2$ perpendicular to $\vec{P}_1$ is λ then value of $\frac{\lambda}{3}$ is

SECTION-2

One or More Than One Correct Type Questions:

45. Let $l_1 = \int_0^{512} \frac{\{x^{1/3}\}}{\sqrt[3]{x^2}}dx$ and $l_2 = \int_0^2 x^2\{x^3\}dx$ (where {.} denotes fractional part of function), then

(A) $l_1 = l_2$ (B) $l_1 = 9l_2$

(C) $l_1 = 12$ (D) $l_2 = 4$

46. Let A and B be two sets such that $n(\cup) = 14$, $n(A \cap B^C) = 2$, $n(A^C \cap B) = 5$ and $n(B^C) = 7$ then which of the following is/are TRUE?

(A) $n(A \cap B) = n(A \cap B^C)$

(B) $n(A \cap B) = n(A^C \cap B)$

(C) $n(A \cup B) = 9$

(D) $n(A \cap B) = 5$

47. If $f(x) = \cos\{[x-3]+\{x-\{x-\{x\}\}\}\}$ for $x \in \left(0, \frac{\pi}{4}\right)$, (where $\{\cdot\}$ = fractional part function and $[\cdot]$ = Greatest integer function), then which of the following is/are **INCORRECT**

(A) $f^{-1}(x) = \frac{\pi}{2} - \sin^{-1}x$

(B) $f^{-1}(x) = \sin^{-1}x$

(C) $f^{-1}(x) = \sin^{-1}\{x\}$

(D) $f^{-1}(x) = \tan^{-1}\left(\frac{\sqrt{1+x}}{x}\right)$

48. For the equation $\frac{40}{x-1} - \frac{160}{x-4} - \frac{200}{x-5} + \frac{320}{x-8} = 6x^2 - 27x$

(A) Number of distinct real solutions of above equation is 3

(B) If E denotes the product of all non-zero roots (real or imaginary) of the equation, then sum of divisors of E is 2904

(C) If S denotes the set of all real roots of equation then, sum of elements of S is 10

(D) If $\alpha_1, \alpha_2 \in R$ be two roots of equation such that $\log_{\alpha_2}(2\alpha_1)$ is defined and can be equals to 1.

49. Let α and β be number of points of discontinuity and non-differentiability respectively of $f(x) = \max.\{x^3, e^x\}$ then

(A) $\alpha = 0$ (B) $\alpha = 2$

(C) $\beta = 0$ (D) $\beta = 2$

50. If $f(x) = 2|x+1| - |x-1| + 3|x-2| + 2x + 1$ and $h(x) = \begin{cases} 2x & ; \quad x \geq 3 \\ 1 & ; \quad x < 3 \end{cases}$, then choose the correct statements

(A) $y = h(f(x))$ is continuous $\forall\, x \in R$
(B) $y = h(f(x))$ is discontinuous at 3 points
(C) $y = h(f(x))$ is not differentiable at 3 points
(D) $y = h(f(x))$ is not differentiable at 4 points

SECTION-3

Single Correct Type Questions:

51. Consider three observations a, b and c such that $b = a + c$. If the standard deviation of $a + 2$, $b + 2$, $c + 2$ is d, then which of the following is true?

(A) $b^2 = 3(a^2 + c^2) + 9d^2$
(B) $b^2 = a^2 + c^2 + 3d^2$
(C) $b^2 = 3(a^2 + c^2 + d^2)$
(D) $b^2 = 3(a^2 + c^2) - 9d^2$

52. Value of $\sum_{r=1}^{10} {}^{15}C_r \cdot {}^{35}C_{10-r}$ is

(A) ${}^{50}C_{10} - {}^{35}C_{15}$ (B) ${}^{50}C_{10} - {}^{35}C_{10}$
(C) ${}^{50}C_{10} - {}^{15}C_{10}$ (D) ${}^{50}C_{10} - {}^{15}C_{15}$

53. If $ax^2 + 2hxy + by^2 = 0$ then $\frac{dy}{dx}$ is

(A) $\frac{y}{x}$ (B) $-\frac{y}{x}$
(C) $\frac{x}{y}$ (D) $-\frac{x}{y}$

54. The value of k such that lines $\frac{x-k}{1} = \frac{y-2}{k} = \frac{z-k}{2}$ and $\vec{r} = \hat{i} + \lambda\left(\hat{i} + \hat{j} + \hat{k}\right)$ are perpendicular is λ then value of $\log_{\left(-\lambda^{-1}\right)}\left(\lambda^2 + 3\lambda - \lambda^3\right)$ is

(A) 1 (B) 5
(C) –5 (D) –3

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